Quadriparesis as an Unusual Manifestation of Hypercalcemia

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ABSTRACT

Quadriparesis as the presenting complaint in hypercalcemia has never been reported. This paper reports two patients in whom quadriparesis was the predominant manifestation of a hypercalcemic state.

Hypercalcemia can produce bizarre, puzzling and confusing symptoms of such a diffuse nature that the patient may be thought to have a functional illness. The most common symptoms include fatigue, weakness, lethargy, constipation, nausea, anorexia and depression. Many patients with mild hypercalcemia are asymptomatic as we have found now that the SMA 12/60 gives us a read out for calcium.

Hypocalcemia is known to cause neuromuscular symptoms as well. The most common symptom is tetany with cramping and muscle irritability. Knapp and Gough[1] have reported three patients with weakness and signs suggestive of spinal cord compression. These patients had signs of posterior cord involvement and urinary incontinence or retention. This type of presentation has not been reported in hypercalcemia but the following cases demonstrate that calcium can be the culprit whether it is too high or too low.

CASE REPORTS

E. S. was a 43-year-old Black female admitted to Metropolitan Hospital, New York Medical College on 10/8/69. Her original admission was to the Urology Service because she entered with the complaint of sudden onset of sharp crampy pain in the suprapubic region with radiation into the left thigh. She complained of weakness in her legs but both the patient and admitting physician attributed this to pain. Other history included amenorrhea for seven months and heavy intake of alcohol for five years. On examination she was obese, lethargic and had "clouded sensorium." BP was 140/90, pulse 108 and regular, temperature was 98.6 (37C). Her neck was supple; her abdomen was obese but except for her neurologic signs the remainder of the physical examination was normal. Neurologically she was found to be unable to move her left lower extremity. There was slight atrophy of the left calf and there was no tenderness on palpation. She had bilateral hyperreflexia, bilateral Babinski's and was incontinent of both urine and feces. A lumbar puncture, myelogram and radionuclide scan were all within normal limits. Her EEG was abnormal and was characterized by generalized 5-6 cps activity. An echoencephalogram showed no mid line shift. It was felt that the patient had a transverse myelitis secondary to a viral infection and was treated with Decadron® over a five-day period. Laboratory studies on admission did not include calcium determinations. The patient improved considerably on steroids and was able to walk and control bladder and bowel function. On 11/3/77, because hepatomegaly had been noted on a follow-up examination, a "liver profile" was performed. Calcium at that time was 13.6 mg%. Over the next few days she again became lethargic and weak. She rapidly progressed to coma and a repeat calcium was 17 mg%. She was treated with phosphates by nasogastric tube and a one dose intravenous solution of sodium sulfate. As her calcium decreased she became symptomatically improved. A cause of her calcium rise was sought but not found. The patient developed a bleeding diathesis and expired two months after admission. An autopsy was refused by the family.

A. P. is a 47-year-old Black female who was well until December of 1971 when she was admitted to Perth Amboy Hospital in New Jersey with severe low back pain. She had a myelogram during that admission and reportedly developed left hemiparesis following the procedure. In May of 1972, she was again admitted to PAH because of headache. She was found to be hypercalcemic and underwent a neck dissection which revealed normal parathyroids. Postoperatively she developed some weakness of the right upper extremity. In August 1972, the patient was admitted because of increasing weakness in both upper and lower extremities. On admission she was found to have normal vital signs. She was diffusely weak and had bilateral sustained ankle clonus. She was found to have an anemia of iron deficiency. Lumbar puncture was performed revealing a protein of 85 mg% with 4 WBC's. Her calcium ranged from 10 to 14 mg. EEG, brain scan and skull x-rays were normal. Because the patient was thought to have a neuromuscular disorder, she was transferred to the Columbia Presbyterian Medical Center, Neurological Institute. On admission at NI, the patient was found to be quadraparetic with a sensory level at approximately C2. The patient was lethargic and had difficulty cooperating. EEG's were repeated, two being reported as normal and the third showing rhythmic slowing. Nerve conduction and electromyographic studies were compatible with a polyneuropathy. A muscle biopsy indicated skeletal muscle atrophy of neurogenic type. A myelogram was essentially normal up to the clivus. Numerous laboratory studies were carried out with positive results including elevation of serum calcium, renal calculus in the upper pole of the right kidney, positive serology (the patient had a history of adequate treatment in the past) and positive PPD. Numerous other studies were not able to identify neoplastic or infectious disease. Her hypercalcemia was treated with intermittent phospho soda by mouth. As her calcium responded to therapy, she showed progressive increase in strength. She also had gradual return of sensation. She lost her Babinski reflexes and regained bowel and bladder control. She retained clonus in the right ankle.

It was felt that she probably had either a parathyroid adenoma or a parathormone secreting neoplasm. No source of neoplasm was located and a neck dissection was recommended but refused by the patient who was discharged to home on phospho soda 10cc TID.

Primary hyperparathyroidism, either as a result of tumor or hyperplasia of the parathyroid glands, is the most common cause of hypercalcemia. Hypervitaminosis D, multiple myelomatosis, metastatic malignancy and milk-alkali syndrome are other less common causes. The advent of automated serum chemical determinations has made us aware of another entity that of asymptomatic hypercalcemia.

The effect of an increase in circulating parathyroid hormone is to increase excretion of phosphorus by the kidney, lowering the serum phosphorus and causing a compensatory rise in calcium

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to maintain ionic balance. The calcium rise is usually at the expense of the bones. Neurological symptoms in chronic cases include headache, weakness and anxiety. A proximal myopathy has been described in hyperparathyroidism, not necessarily in association with hypercalcemia (Vicale[2]). The patient may complain of easy fatigue and muscle cramps and pains. Bischoff and Esslen[3] have described EMG changes suggestive of myopathy without fibrillation potentials to suggest denervation.

Calcium is essential to membrane integrity and function. If calcium concentration is higher than normal there is a decrease in neuromuscular transmission. Hypotonia of all muscles results giving rise to complaints of weakness. Smooth muscle is also involved which would explain constipation and urinary retention. Involvement of cardiac muscle can shorten the QT interval. Hypocalcemia can, in extreme cases cause very similar neuromuscular changes. Initial decrease in calcium concentration causes hyperexcitability but as calcium decreases, transmission of impulses may be blocked completely.

The patients described each had at least one abnormal EEG. EEG changes in hypercalcemia have been reported by several authors including Cohn and Sode[4] and Allen, et al.[5] These changes are usually related to the calcium level, including generalized slowing with excessive amounts of theta and delta ranged activity as well as high voltage bilaterally synchronous slowing. The slow wave activity generally returns to normal sometimes over a period of several weeks and not as a reflection of changes in the serum calcium. Neuromuscular symptoms cleared as the calcium levels declined indicating that there is no structural change in the neuromuscular junctions.

Treatment of the hypercalcemic state includes restriction of calcium intake, hydration, saline infusion, and use of a diuretic-like furosemide. The patient should be kept mobile since immobility increases the calcium loss from bone. Steroids, used almost accidentally in the first case will help to lower serum calcium in patients with myeloma, sarcoidosis, vitamin D intoxication and in some patients with breast carcinoma. A recent addition to treatment available is dialysis. Mithramycin has been recommended especially in patients with hypercalcemia in association with metastatic carcinoma. Sodium sulfate may be given intravenously instead of sodium chloride. Oral phosphates may be useful although in a patient with vomiting, diarrhea, or coma its usefulness is limited. Phosphates in themselves can cause diarrhea.

Effective treatment of hypercalcemia begins with diagnosis. Realizing that hyper or hypocalcemia may present with unusual neurological manifestation including a picture simulating transverse myelitis or a cord lesion, will eliminate time spent in unproductive studies.

References

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A METHOD FOR IMPROVING THE USE OF CONSTANT INFUSION DRUGS IN CARDIOTHORACIC PATIENTS

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to allow one to pick an adequate dose of drug without excessive fluid administration. By the simple expedient of making a drug solution based on the patient's weight (mcg/kg/ml), drug dose tables can be used which indicate the drug dose in micrograms per kilogram per minute (mcg/kg/min) when the hourly infusion rate is known.

Footnote: We will gladly furnish to interested persons copies of the IV rate-drug dose tables together with appropriate weight multiplication factors for dopamine, nitroprusside, nitroglycerin, and epinephrine.

References

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